

GAUSSIAN SIMPLE LINEAR REGRESSION

Assume that on n statistical units we observe (y_i, x_i) , $i=1, \dots, n$.

We assume that each y_i is realization of a random variable Y_i , and that $Y_1, \dots, Y_n$ are independent.

We only consider one covariate x_i , $i=1, \dots, n$.

Consider the model

$$Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i \quad i=1, \dots, n$$

HYPOTHESES:

1. $E[\varepsilon_i] = 0 \quad i=1, \dots, n$
 2. $\text{Var}(\varepsilon_i) = \sigma^2$ for all $i=1, \dots, n$
 3. $\text{Cov}(\varepsilon_i, \varepsilon_k) = 0 \quad i \neq k; \quad i, k=1, \dots, n$
- + 4. ε_i have Gaussian distribution
- $\Rightarrow \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \quad i=1, \dots, n$
- hypo from last time

Consequences for Y_i : we have $Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$

the normal distribution is closed w.r.t. linear transformations

$$\Rightarrow Y_i \sim N(\beta_1 + \beta_2 x_i, \sigma^2) \text{ independent (but not identically distributed)}$$

Now we have distributive assumptions, hence we can derive the estimators for $\beta_1, \beta_2, \sigma^2$ using the maximum likelihood method.

here, the parameters are $(\beta_1, \beta_2, \sigma^2) \rightarrow$ parameter space $\Theta = \mathbb{R}^2 \times (0, +\infty)$

sample space $\mathcal{Y} = \mathbb{R}^n$

Likelihood function $L(\theta) \propto f(y_1, \dots, y_n; \theta) = \prod_{i=1}^n f(y_i; \theta)$

$$\begin{aligned} L(\beta_1, \beta_2, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - \beta_1 - \beta_2 x_i)^2\right\} \\ &= \left(\frac{1}{2\pi}\right)^{\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2\right\} \end{aligned}$$

log-likelihood $\ell(\beta_1, \beta_2, \sigma^2) = \log L(\beta_1, \beta_2, \sigma^2) =$

$$= -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2$$

score function

in general, is the vector $e_\theta(\theta) = \left[\frac{\partial}{\partial \theta_1} \ell(\theta), \dots, \frac{\partial}{\partial \theta_q} \ell(\theta) \right]^T$ where $\theta = (\theta_1, \dots, \theta_q)^T$

here, $q=3 \quad \theta = (\beta_1, \beta_2, \sigma^2)^T$

$$\begin{cases} \frac{\partial}{\partial \beta_1} \ell(\beta_1, \beta_2, \sigma^2) = -\frac{1}{\sigma^2} \sum_{i=1}^n (-1)(y_i - \beta_1 - \beta_2 x_i) = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) \\ \frac{\partial}{\partial \beta_2} \ell(\beta_1, \beta_2, \sigma^2) = -\frac{1}{\sigma^2} \sum_{i=1}^n (-x_i)(y_i - \beta_1 - \beta_2 x_i) = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) x_i \\ \frac{\partial}{\partial \sigma^2} \ell(\beta_1, \beta_2, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2 \end{cases}$$

Likelihood equations

the MLE is found as $\hat{\theta}$ s.t. $e_\theta(\hat{\theta}) = 0$

$$\begin{cases} \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) = 0 & \leadsto \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) = 0 \quad (1) \\ \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) x_i = 0 & \leadsto \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) x_i = 0 \quad (2) \\ -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2 = 0 & (3) \end{cases}$$

(1) and (2) are exactly the same equations we already solved using OLS
they do not depend on σ^2

hence

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} \quad \text{and} \quad \hat{\beta}_2 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

are maximum likelihood estimates.

Let's find the maximum likelihood estimate of σ^2 .

Solving (3)

$$\begin{aligned} -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2 &= 0 \\ -\frac{1}{2(\sigma^2)^2} \left[n\sigma^2 - \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2 \right] &= 0 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{\sum_{i=1}^n (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i)^2}{n} \quad \text{MLE of } \sigma^2 \end{aligned}$$

The matrix of the 2nd derivatives

$$e_{\theta\theta}(\theta) = \left\{ \frac{\partial^2 \ell(\theta)}{\partial \theta_s \partial \theta_r} \right\}_{s,r=1,2,3}$$

$$\begin{aligned} \frac{\partial^2}{\partial \beta_1^2} \ell(\theta) &= -\frac{n}{\sigma^2} & \frac{\partial^2}{\partial \beta_1 \partial \beta_2} \ell(\theta) &= -\frac{n\bar{x}}{\sigma^2} & \frac{\partial^2}{\partial \beta_2 \partial \sigma^2} \ell(\theta) &= -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) \\ \frac{\partial^2}{\partial \beta_2^2} \ell(\theta) &= -\frac{\sum_{i=1}^n x_i^2}{\sigma^2} & \frac{\partial^2}{\partial \beta_2 \partial \sigma^2} \ell(\theta) &= -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n x_i (y_i - \beta_1 - \beta_2 x_i) \\ \frac{\partial^2}{\partial (\sigma^2)^2} \ell(\theta) &= \frac{n}{2(\sigma^2)^2} - \frac{1}{(\sigma^2)^3} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2 \end{aligned}$$

Hence $e_{\theta\theta}(\beta_1, \beta_2, \sigma^2)$ is the matrix

$$\begin{bmatrix} -\frac{n}{\sigma^2} & -\frac{n\bar{x}}{\sigma^2} & -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) \\ -\frac{n\bar{x}}{\sigma^2} & -\frac{\sum_{i=1}^n x_i^2}{\sigma^2} & -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n x_i (y_i - \beta_1 - \beta_2 x_i) \\ -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) & -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n x_i (y_i - \beta_1 - \beta_2 x_i) & \frac{n}{2(\sigma^2)^2} - \frac{1}{(\sigma^2)^3} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2 \end{bmatrix}$$

We need to evaluate these derivatives at $(\hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}^2)$

Both $\frac{\partial^2}{\partial \beta_1 \partial \sigma^2} \ell(\theta)$ and $\frac{\partial^2}{\partial \beta_2 \partial \sigma^2} \ell(\theta)$ are $= 0$ at $(\hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}^2)$.

$$\begin{aligned} \frac{\partial^2}{\partial (\sigma^2)^2} \ell(\theta) \Big|_{\theta=\hat{\theta}} &= \frac{n}{2(\hat{\sigma}^2)^2} - \frac{1}{(\hat{\sigma}^2)^3} \underbrace{\sum_{i=1}^n (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i)^2}_{= n\hat{\sigma}^2} \\ &= \frac{n}{2(\hat{\sigma}^2)^2} - \frac{n}{(\hat{\sigma}^2)^2} \\ &= -\frac{n}{2(\hat{\sigma}^2)^2} \end{aligned}$$

The observed information $j(\hat{\theta}) = -e_{\theta\theta}(\hat{\theta})$ then is

$$j(\hat{\theta}) = \begin{bmatrix} \frac{n}{\hat{\sigma}^2} & \frac{n\bar{x}}{\hat{\sigma}^2} & 0 \\ \frac{n\bar{x}}{\hat{\sigma}^2} & \frac{\sum_{i=1}^n x_i^2}{\hat{\sigma}^2} & 0 \\ 0 & 0 & \frac{n}{2(\hat{\sigma}^2)^2} \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & b \end{bmatrix}$$

and it is possible to show that $(\hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}^2)$ is a maximum.

The maximum likelihood estimates of $(\beta_1, \beta_2, \sigma^2)$ are

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} \quad \text{and}$$

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i)^2}{n}$$

these are the arguments of the lik. equations (1) and (2). Hence they are $= 0$ if evaluated at $(\hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}^2)$